

Q1 for i in {0..10}
do
 echo "\$i"
done

Q2 for i in {0..50}
do
 if [$((i \% 2) == 0)$]; then
 echo \$i
 fi
done

Q3 echo "enter a number: "
read num
i = 1
until [$i -gt 15$]
do
 ~~echo \$(num * i)~~
 echo $$(num * i)$
 i = $$(i + 1)$
done

Q4 ~~read~~ read -ra numbers < nums.txt
echo "The array is: " \${numbers[@]}
for i in "\${numbers[@]}"
do
 echo "The double of the element is: " $$(($i * 2))$
done

Q5

function divide {

local num1 = \$1

local num2 = \$2

if [\$num2 -eq 0]; then

echo "Division by 0 not allowed"

fi

local quotient = \$(echo "scale=2; \$num1/\$num2" | bc)

local remainder = \$(echo "scale=2; \$num1%\$num2" | bc)

echo "The quotient is: " \$quotient

echo "The remainder is: " \$remainder

}

Q6

function maximum {

a = \$1

b = \$2

if [\$a -gt \$b]; then

echo "\$a greater than \$b"

else

echo "\$b greater than \$a"

fi

}

Q7

function directory {

echo "enter a ^{directory} ~~name~~:"

read ~~name~~ directory

if [-e \$directory]; then

echo "directory exist"

ls \$directory

else

mkdir \$new_directory

echo "new directory made"

fi

}

Q bash script that ~~creates~~ takes a word & filename then print the lines where the word appears

```
#!/bin/bash
echo "Enter filename"
read file
echo "Enter a word"
read word
grep "$word" "$file"
```

Q bash script that generates multiplication table for a number from 1 to 20

```
#!/bin/bash
echo "give a number"
read num
i = 1
until [ $i -gt 20 ]
do
    echo $((num * i))
    i = $((i + 1))
done
```

Q bash script to give the current disc usage & if it is greater than 80% print a warning message

```
#!/bin/bash
df > outfile
awk 'NR 21 2 { if ($5 > 80)
    print "warning"
    else
    print "less than 80%"
}' < outfile
```

Q check if a file exist and if it does is it readable and writable

```
#!/bin/bash
```

```
echo "give file name:"
```

```
read file
```

```
if [-e "$file"]; then
```

```
    echo "file exist"
```

```
    if [-r "$file"] && [-w "$file"]; then
```

```
        echo "file is readable & writable"
```

```
    else
```

```
        echo "file is not readable & writable"
```

```
    fi
```

```
else
```

```
    echo "file doesn't exist"
```

```
fi
```

Q script that counts the number of words, lines & characters in a file

```
echo "give a file name"
```

```
read file
```

```
lines = $(wc -l < "$file")
```

```
words = $(wc -w < "$file")
```

```
characters = $(wc -m < "$file")
```

```
echo "no of lines: " $lines
```

```
echo "no of words: " $words
```

```
echo "no of characters: " $characters
```

Q list the contents of the current directory and also the name and size of the largest file

```
#!/bin/bash
```

```
max = 0
```

```
ls -l > outfile
```

```
awk 'NR > 1 {
```

```
    if ($5 -gt max) { then
```

```
        max = $5
```

```
        name = $8
```

```
    }
```

```
}
```

```
echo "largest file is: \"$name\" and the size is \"$max"
```